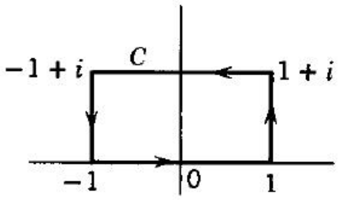
Problem 1 (25 points) Find the integral

$$\oint_C z^2 dz$$

along the following path



$\because z^2$ *在R上解析*

$\therefore$ *根据柯西定理*

$\oint_c z^2 dz = 0$

Problem 2 (25 points): Evaluate the following integral

$$\oint_{|z|=2} \frac{sin\left(\frac{\pi}{2}e^z\right)}{z^3} dz$$

$f'(z) = \frac{\pi}{2}e^z cos\left(\frac{\pi}{2}e^z\right)$

$f''(z) = \frac{\pi^2}{4}e^{2z}\left(cos\left(\frac{\pi}{2}e^z\right) - \sin\left(\frac{\pi}{2}e^z\right)\right)$

$f''(0) = \frac{\pi^2}{4}(0-1) = -\frac{\pi^2}{4}$

$$n = 2 \quad a = 0$$

$$\text{原式} = \frac{2\pi i}{2!} f''(0)$$

$$= \pi i \cdot \left(-\frac{\pi^2}{4}\right)$$

$$= -\frac{\pi^3}{4} i$$

Problem 3 (35 points): Evaluate the following integral:

$$\int_0^{2\pi} e^{2i\theta}\left[\sin\left(e^{i\theta}\right)\right]^2 d\theta$$

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Let $e^{i\theta} = z$

$\therefore \int_0^{2\pi} e^{2i\theta}\left[\sin\left(e^{i\theta}\right)\right]^2 d\theta$

$= \oint_{|z|=1} \frac{z^2[\sin(z)]^2}{iz} dz$

$= -i \oint_{|z|=1} z[\sin(z)]^2 dz$

$\because z[\sin(z)]^2$ is analytic.

$\therefore \int_0^{2\pi} e^{2i\theta}\left[\sin\left(e^{i\theta}\right)\right]^2 d\theta = 0$

Problem 4 (15 points) If $f(z)$ is analytic in region G, and its imaginary part $Im[f(z)] = 0$, prove that $f(z)$ must be a constant inside G.

$\because f(z)$ is analytic in G

$\therefore \oint_G f(z)\, dz = 0$.

即 $\oint_C (u+iv)(dx+i\,dy) = \oint_C (udx - vdy) + i \oint_C (vdx+ud$

$\because v=0$.

~~[struck]~~ $\therefore \oint_C udx + i\oint udy = 0$

$\therefore \oint_C udx = 0$

$\oint_C udy = 0$.

$\therefore$ u与 x, y 均无关 $\therefore$ u为常数, $v=0$

END

$\therefore f(z) = \text{constant}$.